

Metric to Analyze Orbitals in the Active Space

- ▶ Active space selection can be framed as a vector search problem, where the objective is to maximize the overlap between the target molecular orbital coefficients $C_{\text{CASSCF}}^{\text{act}}$ from a converged CASSCF calculation and an initial guess $C_{\text{trial}}^{\text{act}}$.
- ▶ The overlap matrix between these sets of orbitals is defined as:

$$M = (C_{\text{CASSCF}}^{\text{act}})^{\dagger} S (C_{\text{trial}}^{\text{act}}) \quad (1)$$

where S is the atomic orbital overlap matrix.

- ▶ To quantify the similarity between the initial and target orbitals, we perform Singular Value Decomposition (SVD) on S_{overlap} :

$$M = U \Sigma V^{\dagger} \quad (2)$$

where Σ contains the singular values σ_i , which represent the cosines of the angles between the corresponding pairs of orbitals in the two sets.

- ▶ The quality of the active space is then assessed by the Frobenius norm of the matrix of sines of the principal angles, given by:

$$\|\mathbf{d}\|_F = \sqrt{\sum_{i=1}^n (1 - \sigma_i^2)} \quad (3)$$

Minimizing $\|\mathbf{d}\|_F$ thus maximizes the (spatial?) overlap between the initial and target orbitals.

Key Equations

► Mean Squared Error Loss for Orbital Coefficients

$$\lambda = \sum_i (C_i - \hat{C}_i)^2$$

where C_i represents the converged MO coefficient matrices and $\hat{C}_i$ represents the learned MO coefficient matrices for a given geometry i .

► Effective Fock Matrix

$$\hat{F}_{pq} = h_{pq} + \sum_{rs} D_{rs} ((pq|rs) - (qr|ps))$$

where h_{pq} is the core Hamiltonian, D_{rs} is the state-averaged density matrix, and $(pq|rs)$, $(qr|ps)$ are two-electron integrals.

► Symmetrized Learned Effective Fock Matrix

$$\hat{F} = \frac{1}{2}(X + X^T)$$

where X is a learned square matrix, symmetrized to obtain the effective Fock matrix $\hat{F}$.

Reference for the number of virtual orbitals

Basis Set	NBF	NVO
STO-3G	2	0
3-21G	4	2
6-31G	4	2
cc-pVDZ	10	8
cc-pVTZ	28	26
aug-cc-pVDZ	18	16
aug-cc-pVTZ	46	44

Table: Number of Basis Functions (NBF) and Virtual Orbitals (NVO) for H_2 in Various Basis Sets

Example: $\|\mathbf{d}\|_F$ of H_2 CAS(2,2) orbitals at equilibrium

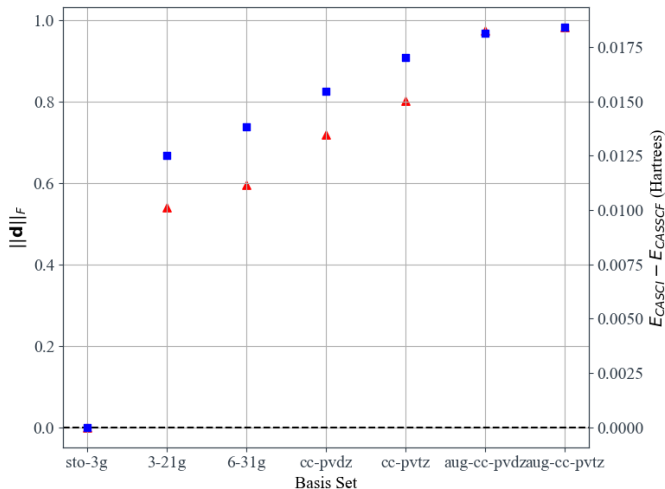


Figure: $\|\mathbf{d}\|_F$ (red) and $E_{\text{CASCI}} - E_{\text{CASSCF}}$ (blue) as a function of basis set size

Optimizing the Orbitals within an Active Space

- ▶ The optimization problem for selecting the right orbitals in an active space involves minimizing the Frobenius norm of the matrix of sines of principal angles under the constraint of orthonormal molecular orbitals:

$$\min_{C_{\text{trial}}^{\text{act}}} ||\mathbf{d}||_F \quad \text{subject to} \quad (C_{\text{trial}}^{\text{act}})^{\dagger} S (C_{\text{trial}}^{\text{act}}) = I \quad (4)$$

- ▶ **Gradient-Based Optimization:**

- ▶ The update equation for the molecular orbital coefficients $C_{\text{trial}}^{\text{act}}$ at iteration k is:

$$C_{\text{trial}}^{\text{act},k+1} = C_{\text{trial}}^{\text{act},k} - \alpha_k \nabla ||\mathbf{d}||_F^k \quad (5)$$

where α_k is the step size, and $\nabla ||\mathbf{d}||_F^k$ is the gradient of the Frobenius norm at iteration k .

- ▶ **Automatic Differentiation (Autodiff) Based Optimization:**

- ▶ The optimization step is similar to the gradient-based method but utilizes autodiff for gradient computation:

$$C_{\text{trial}}^{\text{act},k+1} = \text{optimizer_step}(C_{\text{trial}}^{\text{act},k}, \nabla ||\mathbf{d}||_F^k) \quad (6)$$

where $\nabla ||\mathbf{d}||_F^k$ is the gradient computed via autodiff.

- ▶ In both techniques, maintaining the orthonormality of $C_{\text{trial}}^{\text{act}}$ can be achieved through Gram-Schmidt after each update.

Optimization Algorithm

Objective Function

Let C denote the trial coefficient matrix, C_{CASSCF} the CASSCF coefficient matrix, and S the overlap matrix. The objective function is defined as:

$$f_1(C) = \|\sin(\Sigma(C, C_{\text{CASSCF}}, S))\|_F$$

where Σ is the singular value matrix and $\|\cdot\|_F$ denotes the Frobenius norm.

Gradient Descent

The gradient descent step is given by:

$$C_{\text{trial}}^{(k+1)} = C_{\text{trial}}^{(k)} - \eta \nabla f(C_{\text{trial}}^{(k)})$$

where η is the learning rate.

Re-orthonormalization

After each gradient update, re-orthonormalize C_{trial} .

Algorithm Steps

Convergence Criteria

The algorithm stops when $\|\mathbf{d}\|_k < \epsilon$, a predetermined threshold.

Algorithm Steps

1. Initialize $\mathbf{C}^{(0)}$ with RHF/UHF orbitals.
2. Compute the gradient $\nabla f(\mathbf{C}^{(k)})$.
3. Update $\mathbf{C}^{(k)}$ using the gradient descent step.
4. Check the convergence criteria.
5. If the criteria are not met, return to step 2.

Gradient Descent and Reorthonormalization

Gradient Descent to maximize overlap

The gradient descent step now minimizes the modified objective function:

$$C_{\text{trial}}^{(k+1)} = C_{\text{trial}}^{(k)} - \eta \nabla f(C_{\text{trial}}^{(k)})$$

where η is the learning rate and $\nabla f(C)$ includes the gradient of the energy difference term.

Re-orthonormalization

After each gradient update, re-orthonormalize C_{trial} .

Example: H_2 at equilibrium in 3-21G

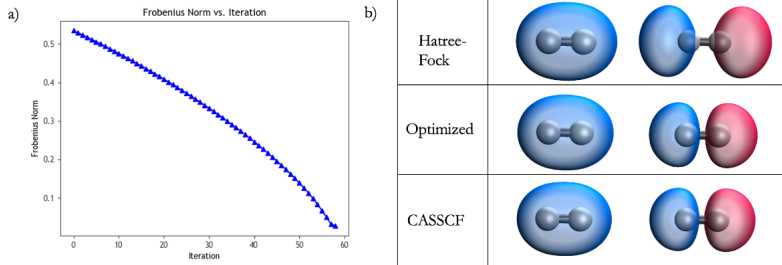


Figure: a) $\|\mathbf{d}\|_F$ as a function of iteration steps, b) Visualization of Initial, Final, and Target CASSCF orbitals

	Hartree-Fock	CASCI (C_{trial})	CASCI (C_{opt})	CASSCF
Energy (a.u)	-1.122940	-1.129889	-1.140710	-1.142407

Table: Comparison of computed energies with different orbitals. The descent pathway was truncated with a convergence criterion $\|\mathbf{d}\|_F \leq 0.03$

Including Energy Difference in Optimization

Energy Criterion in the Objective Function

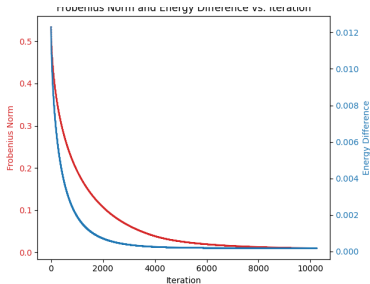
To include the energy difference between CASCI and CASSCF energies, the objective function is modified as:

$$f_2(C) = \|\sin(\Sigma(C, C_{\text{CASSCF}}, S))\|_F + |E_{\text{CASCI}}(C) - E_{\text{CASSCF}}| + \lambda_{\text{reg}} \|C_{\text{trial}} - C_{\text{CASSCF}}\|^2$$

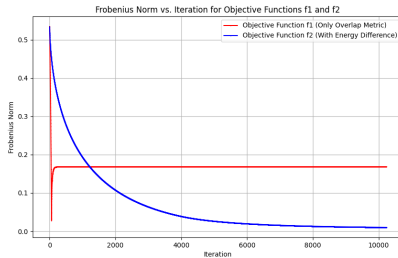
where:

- ▶ Σ is the singular value matrix.
- ▶ λ_{reg} is a regularization parameter.
- ▶ $E_{\text{CASCI}}(C)$ is the CASCI energy computed with C .
- ▶ E_{CASSCF} is the CASSCF energy with optimized coefficients.

Frobenius Norm and Energy Difference as a Function of Iterations: H2 in 3-21G



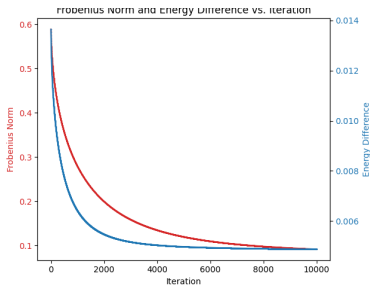
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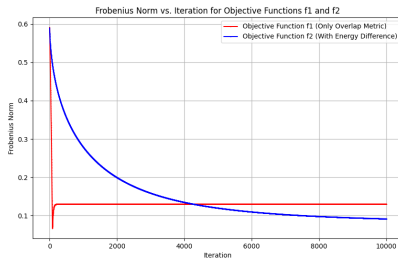
b

Figure: a) Frobenius norm of the matrix of sines ($|\mathbf{d}|_F$) whose value determines the overlap between the trial and optimized MO coefficient matrix plotted along with $|E_{\text{CASSCI}}(C) - E_{\text{CASSCF}}|$ as a function of iterations. b) $|\mathbf{d}|_F$ for the objective function f_1 (only overlap metric) and f_2 (overlap and energy) as a function of iteration.

Frobenius Norm and Energy Difference as a Function of Iterations: H2 in 6-31G



a



b

Figure: a) Frobenius norm of the matrix of sines ($|\mathbf{d}|_F$) whose value determines the overlap between the trial and optimized MO coefficient matrix plotted along with $|E_{\text{CASSCI}}(C) - E_{\text{CASSCF}}|$ as a function of iterations. b) $|\mathbf{d}|_F$ for the objective function f_1 (only overlap metric) and f_2 (overlap and energy) as a function of iteration.

Orbital Optimizations and their thresholds: H2 in 3-21G

Orbitals	$E_{\text{CASCI}}(C)$	dE (a.u)	dE (kcal/mol)
Hartree-Fock	-1.12989	0.01252	7.86
f_1 ($ \mathbf{d} _F \leq 0.03$)	-1.14071	0.00167	1.06
f_1 ($ \mathbf{d} _F \leq 0.01$)	-1.13961	0.00280	1.76
f_2 ($ \mathbf{d} _F \leq 0.01$)	-1.14223	0.00018	0.11

Table: Deviations of dE of CASCI energies from CASSCF energies for MO coefficients optimized with various schemes, starting from Hartree-Fock, and then gradient descent with an objective function f_1 with two different convergence thresholds and with an objective function f_2 . f_1 contains only an orbital overlap metric, whereas f_2 contains both an energy and overlap metric.

CASCI Wave Function and Exponential Parametrization

► CASCI Wave Function Representation:

$$|\Psi_J\rangle = \sum_I C_{IJ} |\Phi_I\rangle \quad (7)$$

This equation represents the Complete Active Space Configuration Interaction (CASCI) wave function for state J as a linear combination of Slater determinants $|\Phi_I\rangle$. The coefficients C_{IJ} are the configuration interaction (CI) expansion coefficients.

► Exponential Parametrization of the Wave Function:

$$|\tilde{\Psi}_J\rangle = e^{\hat{R}} e^{\hat{S}} |\Psi_J\rangle \quad (8)$$

This parametrization expresses the wave function $|\Psi_J\rangle$ using exponential operators $e^{\hat{R}}$ and $e^{\hat{S}}$, where R and S represent variations in the orbital and CI coefficients, respectively.

Operators and Antisymmetrization

► Orbital Rotation Operator:

$$\hat{R} = \sum_{p>q} R_{pq} E_{pq} \quad (9)$$

The operator R performs orbital rotations and is expressed as a sum of anti-Hermitian singlet excitation operators E_{pq} , with R_{pq} being the rotation parameters.

► Proper Form of E_{pq} :

$$E_{pq} = a_p^\dagger a_q - a_q^\dagger a_p \quad (10)$$

This antisymmetrized form ensures the operator is anti-Hermitian.

► CI Coefficient Transformation Operator:

$$S = \sum_{K \neq J} S_{KJ} (|\Psi_K\rangle \langle \Psi_J| - |\Psi_J\rangle \langle \Psi_K|) \quad (11)$$

This antisymmetrized form ensures proper contributions from the target state $|\Psi_J\rangle$ and orthogonal states $|\Psi_K\rangle$.

Formulation of the Generalized Fock Matrix

► Generalized Fock Matrix ($\hat{F}$):

$$\hat{F}_{pq} = h_{pq} + \sum_{rs} D_{rs} \left((pq|rs) - \frac{1}{2}(pr|qs) \right)$$

where:

- h_{pq} is the core Hamiltonian.
 - D_{rs} is the density matrix.
 - $(pq|rs)$ and $(pr|qs)$ are two-electron integrals.
- ## ► Orbital Rotation Parameters:

$$C' = C \exp(\kappa)$$

with κ being an anti-symmetric matrix.

Orbital Gradients from the Generalized Fock Matrix

► Energy Gradient:

$$\frac{\partial E}{\partial \kappa_{pq}} = 2\hat{F}_{pq}$$

► Steps to Compute Orbital Gradients:

1. Compute the core Hamiltonian (h_{pq}).
2. Compute the density matrix (D_{rs}).
3. Compute the two-electron integrals.
4. Form the generalized Fock matrix.
5. Calculate orbital gradients using $2\hat{F}_{pq}$.